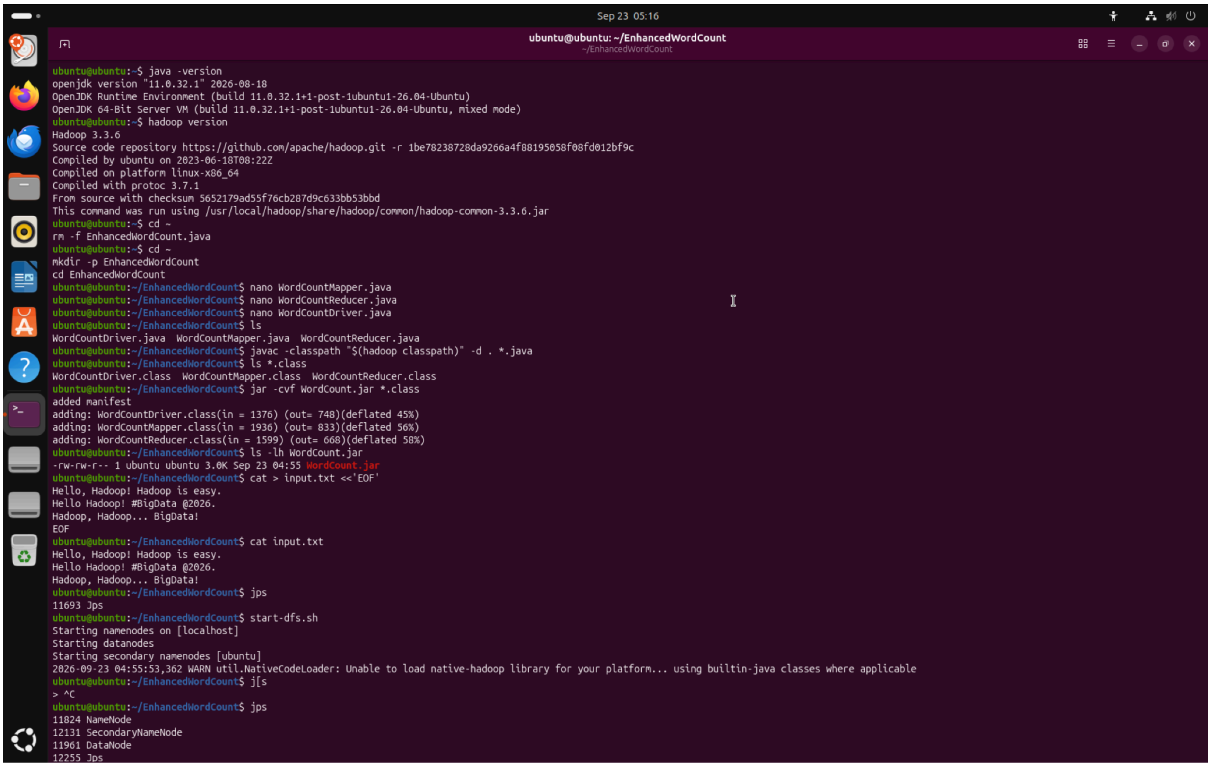
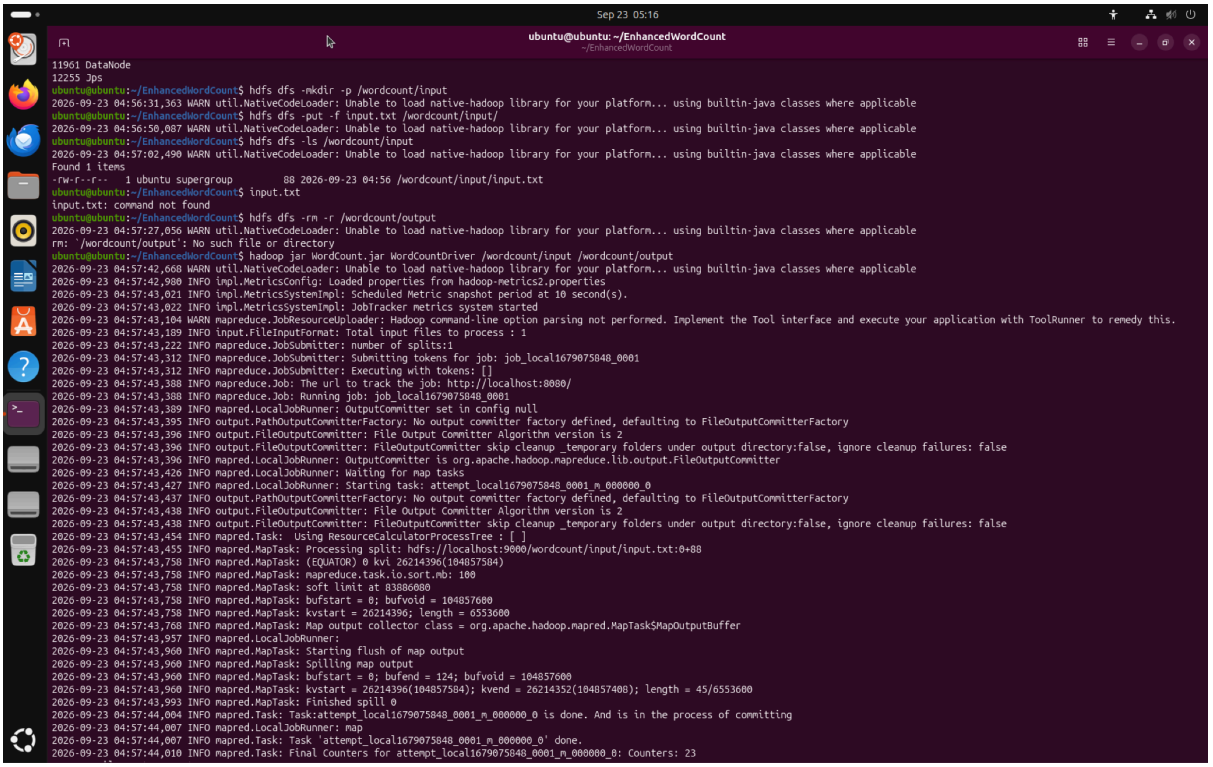
NAME : Adesh Deepak Patankar
CLASS : TY-BDCE

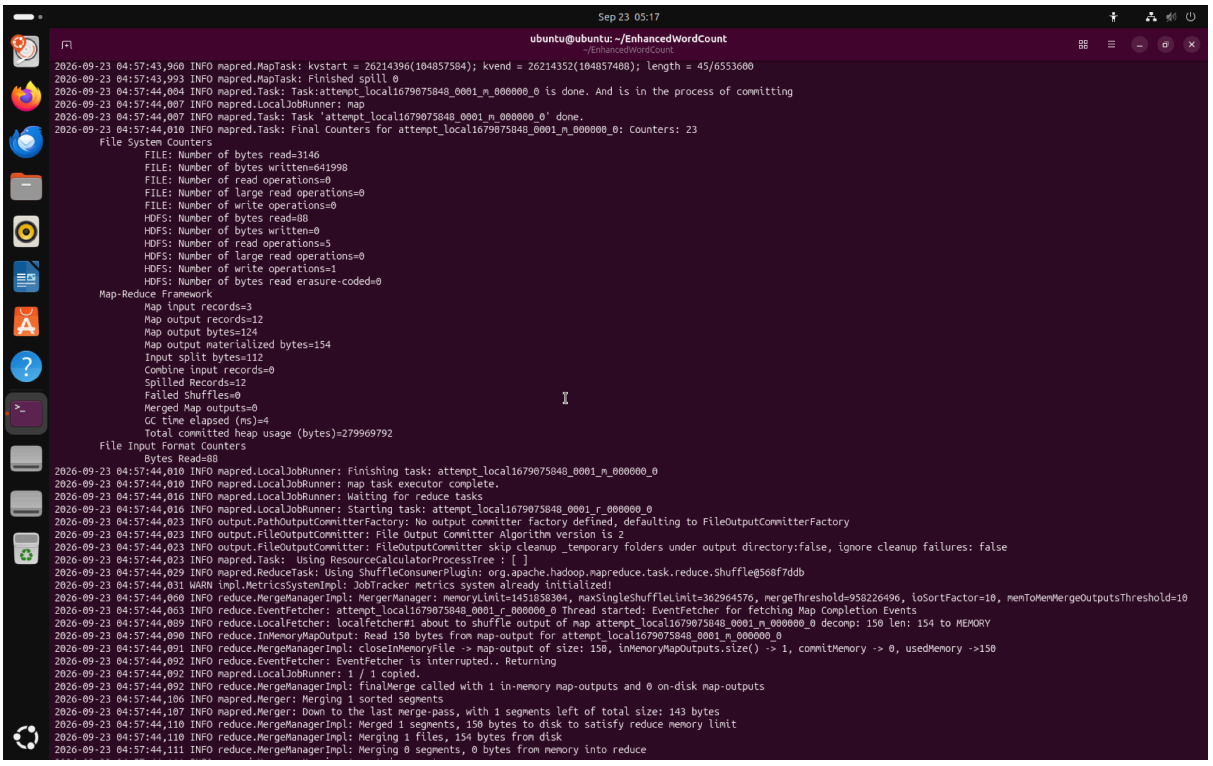
SUBJECT : BDTL - LAB
PBA - 1

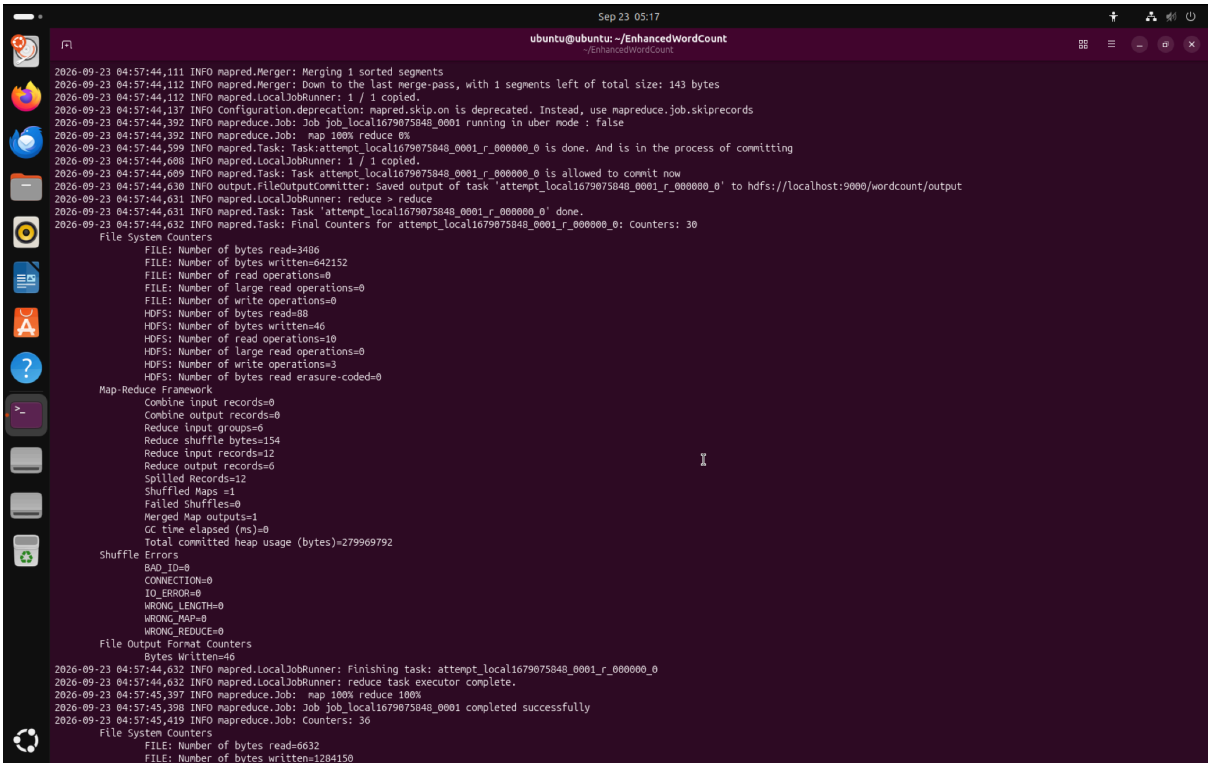
Problem Statement :- Enhance WordCount to exclude punctuation marks and special characters.

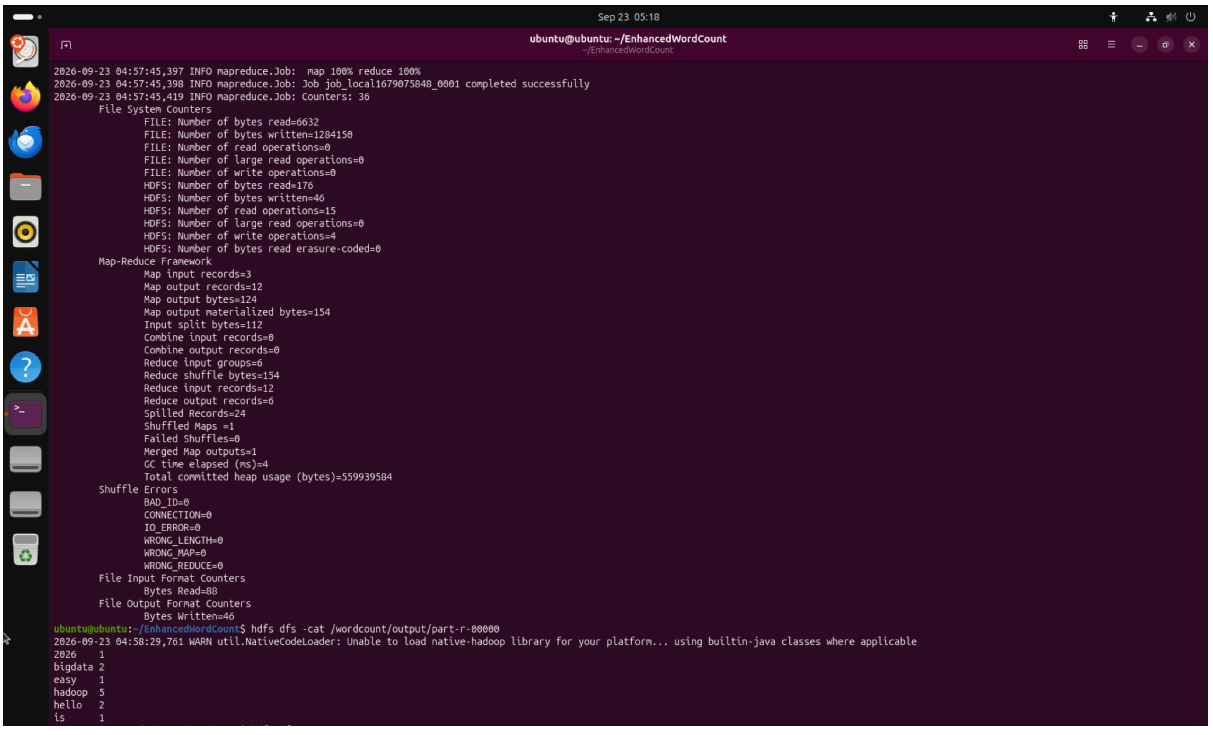
SCREENSHOTS OF THE PERFORMED EXPERIMENT




Sep 23 05:16
ubuntu@ubuntu: ~/EnhancedWordCount
~/EnhancedWordCount
11961 DataNode
12255 Jps
ubuntu@ubuntu:~/EnhancedWordCount$ hdfs dfs -mkdir -p /wordcount/input
2026-09-23 04:56:31,363 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
ubuntu@ubuntu:~/EnhancedWordCount$ hdfs dfs -put -f input.txt /wordcount/input/
2026-09-23 04:56:50,007 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
ubuntu@ubuntu:~/EnhancedWordCount$ hdfs dfs -ls /wordcount/input
2026-09-23 04:57:02,490 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 1 items
-rw-r--r--   1 ubuntu supergroup   88 2026-09-23 04:56 /wordcount/input/input.txt
ubuntu@ubuntu:~/EnhancedWordCount$ input.txt
input.txt: command not found
ubuntu@ubuntu:~/EnhancedWordCount$ hdfs dfs -rm -r /wordcount/output
2026-09-23 04:57:27,056 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
rm: `/wordcount/output': No such file or directory
ubuntu@ubuntu:~/EnhancedWordCount$ hadoop jar WordCount.jar WordCountDriver /wordcount/input /wordcount/output
2026-09-23 04:57:42,660 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-09-23 04:57:42,980 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-09-23 04:57:43,021 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-09-23 04:57:43,022 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-09-23 04:57:43,104 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-23 04:57:43,189 INFO input.FileInputFormat: Total input files to process : 1
2026-09-23 04:57:43,222 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-23 04:57:43,312 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1679075848_0001
2026-09-23 04:57:43,312 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-23 04:57:43,388 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-09-23 04:57:43,388 INFO mapreduce.Job: Running job: job_local1679075848_0001
2026-09-23 04:57:43,389 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-09-23 04:57:43,395 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-09-23 04:57:43,396 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-09-23 04:57:43,396 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-09-23 04:57:43,396 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter
2026-09-23 04:57:43,426 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-09-23 04:57:43,427 INFO mapred.LocalJobRunner: Starting task: attempt_local1679075848_0001_m_000000_0
2026-09-23 04:57:43,437 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-09-23 04:57:43,438 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-09-23 04:57:43,438 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-09-23 04:57:43,454 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-09-23 04:57:43,455 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/wordcount/input/input.txt:0+88
2026-09-23 04:57:43,758 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-09-23 04:57:43,758 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-09-23 04:57:43,758 INFO mapred.MapTask: soft limit at 83886080
2026-09-23 04:57:43,758 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-09-23 04:57:43,758 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-09-23 04:57:43,768 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-09-23 04:57:43,957 INFO mapred.LocalJobRunner:
2026-09-23 04:57:43,960 INFO mapred.MapTask: Starting flush of map output
2026-09-23 04:57:43,960 INFO mapred.MapTask: Spilling map output
2026-09-23 04:57:43,960 INFO mapred.MapTask: bufstart = 0; bufend = 124; bufvoid = 104857600
2026-09-23 04:57:43,960 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214352(104857408); length = 45/6553600
2026-09-23 04:57:43,993 INFO mapred.MapTask: Finished spill 0
2026-09-23 04:57:44,004 INFO mapred.Task: Task:attempt_local1679075848_0001_m_000000_0 is done. And is in the process of committing
2026-09-23 04:57:44,007 INFO mapred.LocalJobRunner: map
2026-09-23 04:57:44,007 INFO mapred.Task: Task 'attempt_local1679075848_0001_m_000000_0' done.
2026-09-23 04:57:44,010 INFO mapred.Task: Final Counters for attempt_local1679075848_0001_m_000000_0: Counters: 23


Sep 23 05:17
ubuntu@ubuntu: ~/EnhancedWordCount
~/EnhancedWordCount
2026-09-23 04:57:43,960 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214352(104857408); length = 45/6553600
2026-09-23 04:57:43,993 INFO mapred.MapTask: Finished spill 0
2026-09-23 04:57:44,004 INFO mapred.Task: Task:attempt_local1679075848_0001_m_000000_0 is done. And is in the process of committing
2026-09-23 04:57:44,007 INFO mapred.LocalJobRunner: map
2026-09-23 04:57:44,007 INFO mapred.Task: Task 'attempt_local1679075848_0001_m_000000_0' done.
2026-09-23 04:57:44,010 INFO mapred.Task: Final Counters for attempt_local1679075848_0001_m_000000_0: Counters: 23
File System Counters
FILE: Number of bytes read=3146
FILE: Number of bytes written=641998
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=88
HDFS: Number of bytes written=0
HDFS: Number of read operations=5
HDFS: Number of large read operations=0
HDFS: Number of write operations=1
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=3
Map output records=12
Map output bytes=124
Map output materialized bytes=154
Input split bytes=112
Combine input records=0
Spilled Records=12
Failed Shuffles=0
Merged Map outputs=0
GC time elapsed (ms)=4
Total committed heap usage (bytes)=279969792
File Input Format Counters
Bytes Read=88
2026-09-23 04:57:44,010 INFO mapred.LocalJobRunner: Finishing task: attempt_local1679075848_0001_m_000000_0
2026-09-23 04:57:44,010 INFO mapred.LocalJobRunner: map task executor complete.
2026-09-23 04:57:44,016 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-09-23 04:57:44,016 INFO mapred.LocalJobRunner: Starting task: attempt_local1679075848_0001_r_000000_0
2026-09-23 04:57:44,023 INFO output.PathOutputCommitterFactory: No output committer factory defined, defaulting to FileOutputCommitterFactory
2026-09-23 04:57:44,023 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-09-23 04:57:44,023 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-09-23 04:57:44,023 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-09-23 04:57:44,029 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@568f7ddb
2026-09-23 04:57:44,031 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-09-23 04:57:44,060 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=1451858304, maxSingleShuffleLimit=362964576, mergeThreshold=958226496, ioSortFactor=10, memToMemMergeOutputsThreshold=10
2026-09-23 04:57:44,063 INFO reduce.EventFetcher: attempt_local1679075848_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-09-23 04:57:44,089 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local1679075848_0001_m_000000_0 decomp: 150 len: 154 to MEMORY
2026-09-23 04:57:44,090 INFO reduce.InMemoryMapOutput: Read 150 bytes from map-output for attempt_local1679075848_0001_m_000000_0
2026-09-23 04:57:44,091 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 150, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory ->150
2026-09-23 04:57:44,092 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-09-23 04:57:44,092 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-09-23 04:57:44,092 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-09-23 04:57:44,106 INFO mapred.Merger: Merging 1 sorted segments
2026-09-23 04:57:44,107 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 143 bytes
2026-09-23 04:57:44,110 INFO reduce.MergeManagerImpl: Merged 1 segments, 150 bytes to disk to satisfy reduce memory limit
2026-09-23 04:57:44,110 INFO reduce.MergeManagerImpl: Merging 1 files, 154 bytes from disk
2026-09-23 04:57:44,111 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce


Sep 23 05:17
ubuntu@ubuntu: ~/EnhancedWordCount
~/EnhancedWordCount
2026-09-23 04:57:44,111 INFO mapred.Merger: Merging 1 sorted segments
2026-09-23 04:57:44,112 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 143 bytes
2026-09-23 04:57:44,112 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-09-23 04:57:44,137 INFO Configuration.deprecation: mapred.skip.on is deprecated. Instead, use mapreduce.job.skiprecords
2026-09-23 04:57:44,392 INFO mapreduce.Job: Job job_local1679075848_0001 running in uber mode : false
2026-09-23 04:57:44,392 INFO mapreduce.Job: map 100% reduce 0%
2026-09-23 04:57:44,599 INFO mapred.Task: Task:attempt_local1679075848_0001_r_000000_0 is done. And is in the process of committing
2026-09-23 04:57:44,608 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-09-23 04:57:44,609 INFO mapred.Task: Task attempt_local1679075848_0001_r_000000_0 is allowed to commit now
2026-09-23 04:57:44,630 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1679075848_0001_r_000000_0' to hdfs://localhost:9000/wordcount/output
2026-09-23 04:57:44,631 INFO mapred.LocalJobRunner: reduce > reduce
2026-09-23 04:57:44,631 INFO mapred.Task: Task 'attempt_local1679075848_0001_r_000000_0' done.
2026-09-23 04:57:44,632 INFO mapred.Task: Final Counters for attempt_local1679075848_0001_r_000000_0: Counters: 30
File System Counters
FILE: Number of bytes read=3486
FILE: Number of bytes written=642152
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=88
HDFS: Number of bytes written=46
HDFS: Number of read operations=10
HDFS: Number of large read operations=0
HDFS: Number of write operations=3
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Combine input records=0
Combine output records=0
Reduce input groups=6
Reduce shuffle bytes=154
Reduce input records=12
Reduce output records=6
Spilled Records=12
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=0
Total committed heap usage (bytes)=279969792
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Output Format Counters
Bytes Written=46
2026-09-23 04:57:44,632 INFO mapred.LocalJobRunner: Finishing task: attempt_local1679075848_0001_r_000000_0
2026-09-23 04:57:44,632 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-09-23 04:57:45,397 INFO mapreduce.Job: map 100% reduce 100%
2026-09-23 04:57:45,398 INFO mapreduce.Job: Job job_local1679075848_0001 completed successfully
2026-09-23 04:57:45,419 INFO mapreduce.Job: Counters: 36
File System Counters
FILE: Number of bytes read=6632
FILE: Number of bytes written=1284150


Sep 23 05:18
ubuntu@ubuntu: ~/EnhancedWordCount
~/EnhancedWordCount
2026-09-23 04:57:45,397 INFO mapreduce.Job: map 100% reduce 100%
2026-09-23 04:57:45,398 INFO mapreduce.Job: Job job_local1679075848_0001 completed successfully
2026-09-23 04:57:45,419 INFO mapreduce.Job: Counters: 36
File System Counters
FILE: Number of bytes read=6632
FILE: Number of bytes written=1284150
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=176
HDFS: Number of bytes written=46
HDFS: Number of read operations=15
HDFS: Number of large read operations=0
HDFS: Number of write operations=4
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
Map input records=3
Map output records=12
Map output bytes=124
Map output materialized bytes=154
Input split bytes=112
Combine input records=0
Combine output records=0
Reduce input groups=6
Reduce shuffle bytes=154
Reduce input records=12
Reduce output records=6
Spilled Records=24
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=4
Total committed heap usage (bytes)=559939584
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=88
File Output Format Counters
Bytes Written=46
ubuntu@ubuntu:~/EnhancedWordCount$ hdfs dfs -cat /wordcount/output/part-r-00000
2026-09-23 04:58:29,761 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026 1
bigdata 2
easy 1
hadoop 5
hello 2
is 1